Question			Answer	Marks	AO	Guidance	
11	(a)		Using $s=\frac{1}{2}(u+v)t$ with $u=0, v=9, t=5$ $s=\frac{1}{2}(0+9)5=22.5$ m	M1 A1* [2]	3.4 1.1b	Allow for any sequence of *suvat* equations leading to a value for *s* ($a=1.8$ may be seen)	
11	(b)		Substituting $t=5, v=9$ into $v=0.05t^3+kt$ $9=0.05\times5^3+5k$ $k=0.55$	M1 A1 [2]	3.3 1.1b	Uses $t=5, v=9$ to form an equation for *k*	
11	(c)		$a=\frac{dv}{dt}=0.15t^2+k$ When $t=5, a=0.15\times5^2+k=4.3$ m s^{-2}	M1 A1 [2]	3.4 1.1b	Differentiation of their model B FT their *k*	
11	(d)		$\int_0^5 v\,dt=\int_0^5(0.05t^3+0.55t)\,dt$ distance 14.6875 m (14.7 m) Model B distance 14.7 is closer to 16 m than model A (22.5 m) so B models this better	M1 A1* E1 (dep) [3]	3.4 1.1a 3.5a	Definite integral oe FT their *k* BC acceptable Comment which quotes at least two values. Dependent on both A marks indicated *	Do not allow for $s=0.05\frac{t^4}{4}+0.55\frac{t^2}{2}+c$ unless limits used, or there is an attempt to evaluate *c*